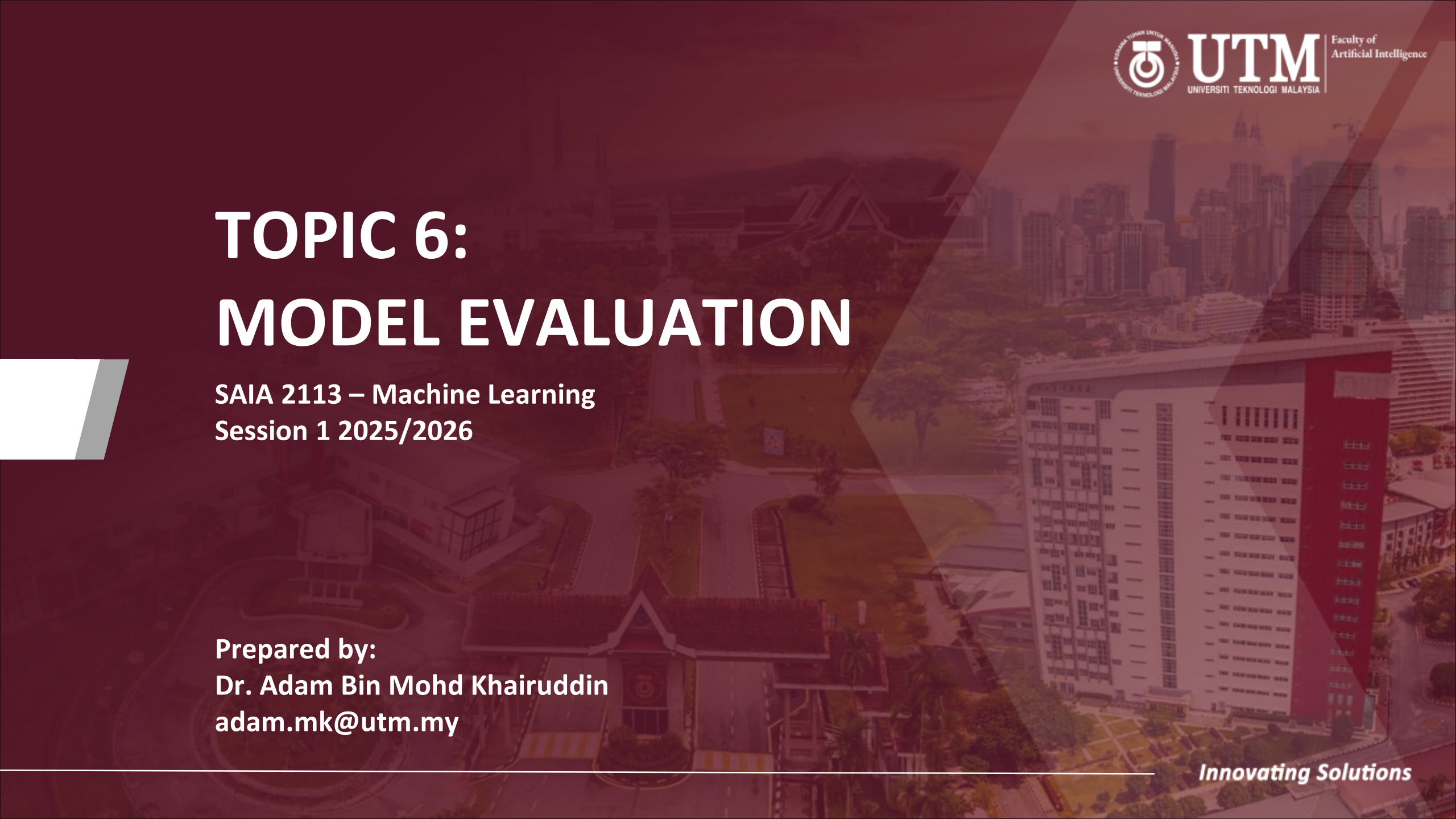


UTM Faculty of Artificial Intelligence
UNIVERSITI TEKNOLOGI MALAYSIA

TOPIC 6:
MODEL EVALUATION

SAIA 2113 – Machine Learning
Session 1 2025/2026

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Innovating Solutions

Topic 6: Model Evaluation

- What is model evaluation?
- Confusion matrix
- Classification: Evaluation metrics
- Classification: ROC, AUC and PRC
- Prediction bias
- Cross validation
- Training, validation, and test sets

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What is model evaluation?

- Model evaluation is the process of utilizing various evaluation metrics to enhance and assess an ML model's performance.
- The performance of the model should be accessed not only during the development phase but also after model deployment phase.
- There are several key metrics that you can use to evaluate your model: (1) accuracy; (2) Precision; (3) Recall (Sensitivity); (4) F1-score; and (5) ROC-AUC.

https://www.coursera.org/articles/model-evaluation

Benefits of model evaluation

Some common benefits of evaluation ML models include:

- Overfitting detection: Evaluating models can help you detect and minimize failures when you integrate new data into the model.
- Improve model performance: Implementing model evaluation and performance metrics eagles the model to gain more insight in order to improve its performance.
- Ongoing improvements: You can reevaluate the model after it is deployed. This allows continuous improvement and model updates.
- Enhanced predictions: After you have evaluated a model, you can use it to make accurate and reliable predictions based on the data it is trained on.

https://www.coursera.org/articles/model-evaluation

Common challenges in model evaluation

Model evaluation may also pose some challenges:

- Data dependency: Dependency on reliable data may be a challenge because you must ensure that you have high-quality, well-labeled data to properly train and evaluate your ML model.
- Incorrect metric selection: If you choose the wrong metrics when performing model evaluation, your ML model may develop incorrect outputs.
- Extensive resource allocations: Model evaluation requires various resources, which may be time-consuming to allocate, and metrics such as cross-validation are very time-intensive.
- Model drift: This process refers to changes in the distribution of data, which can make the initial evaluations you made irrelevant and inaccurate.

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Confusion matrix

The probability score is not reality, or ground truth. There are four possible outcomes for each output from a binary classifier. For the spam classifier example, if you lay out the ground truth as columns and the model's prediction as rows in the following table, called a confusion matrix.

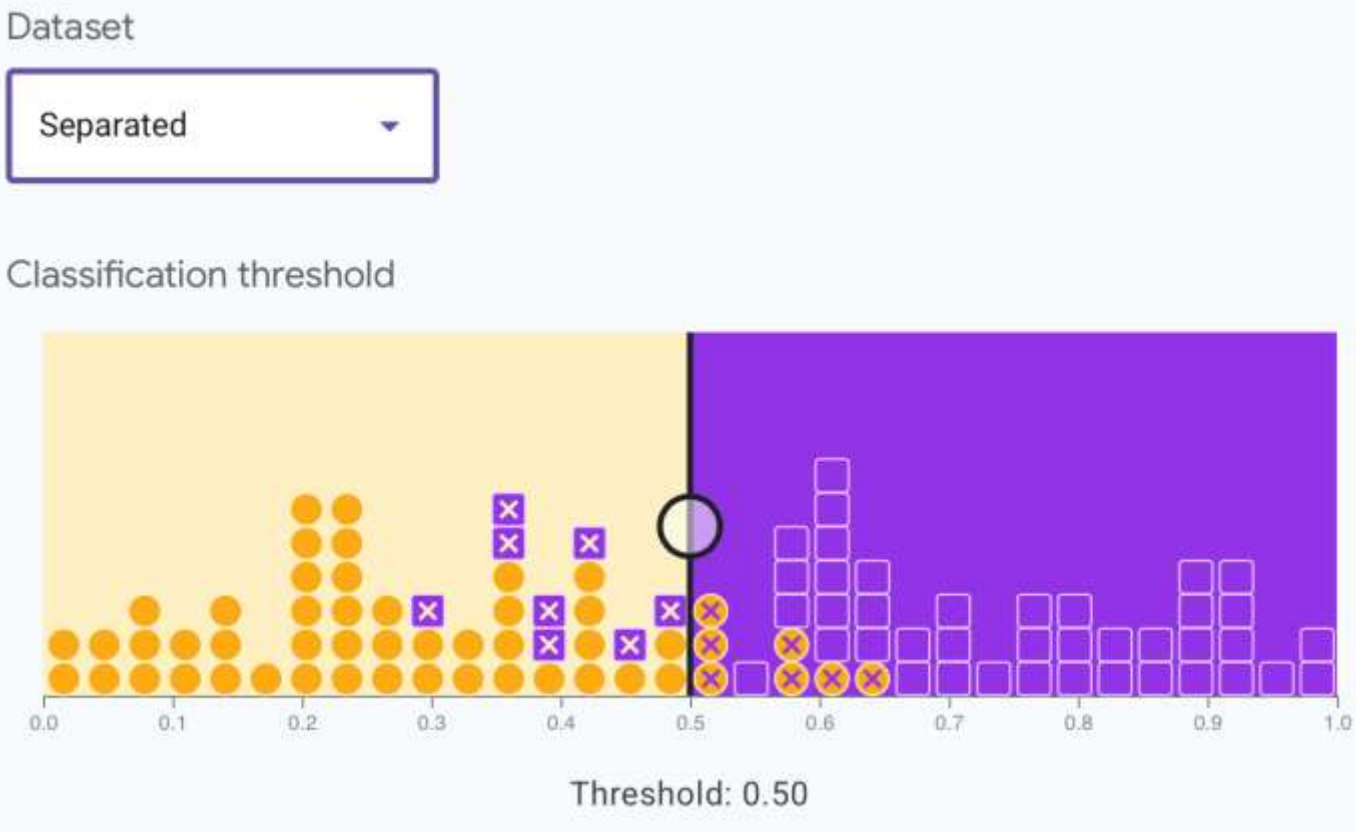
	Actual Positive	**Actual Negative**
Predicted positive	True positive (TP): A spam email correctly classified as a spam email. These are the spam messages automatically sent to the spam folder.	False positive (FP): A not-spam email misclassified as spam. These are the legitimate emails that wind up in the spam folder.
Predicted negative	False negative (FN): A spam email misclassified as not-spam. These are spam emails that aren't caught by the spam filter and make their way into the inbox.	True negative (TN): A not-spam email correctly classified as not-spam. These are the legitimate emails that are sent directly to the inbox.

The total in each row gives all predicted positives (TP + FP) and all predicted negatives (FN + TN), regardless of validity.

The total in each column gives all real positives (TP + FN) and all real negatives (FP + TN) regardless of model classification.

https://developers.google.com/machine-learning/crash-course/classification/thresholding

Confusion matrix



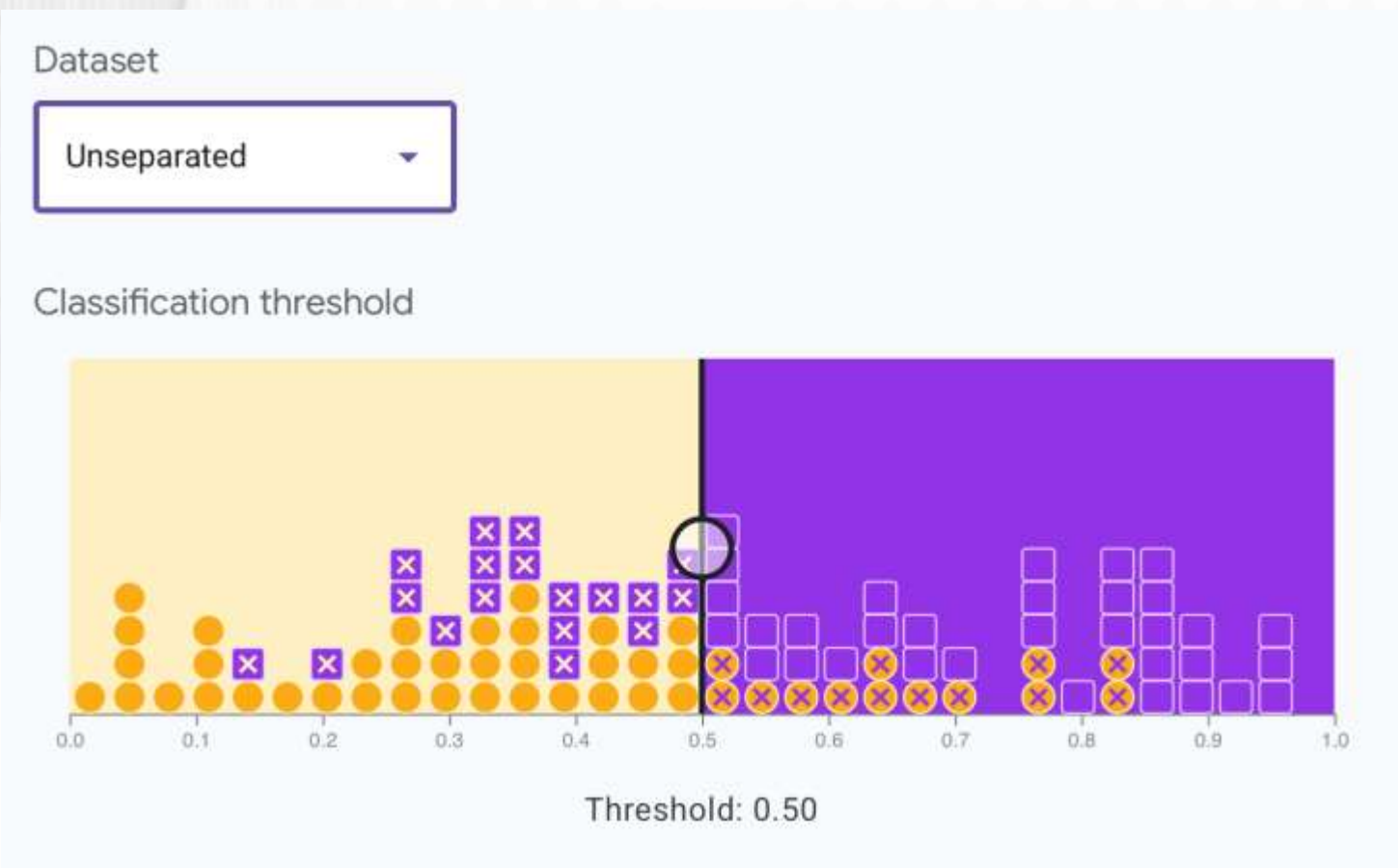
Separated dataset is where positive examples and negative examples are generally well differentiated, with most positive examples having higher scores than negative examples.

Confusion matrix

	Actually positive	Actually negative
Predicted positive	TP=40	FP=7
Predicted negative	FN=8	TN=44

https://developers.google.com/machine-learning/crash-course/classification/thresholding

Confusion matrix



Unseparated dataset is where many positive examples have lower scores than negative examples, and many negative examples have higher scores than positive examples.

Confusion matrix

	Actually positive	Actually negative
Predicted positive	TP=32	FP=12
Predicted negative	FN=19	TN=36

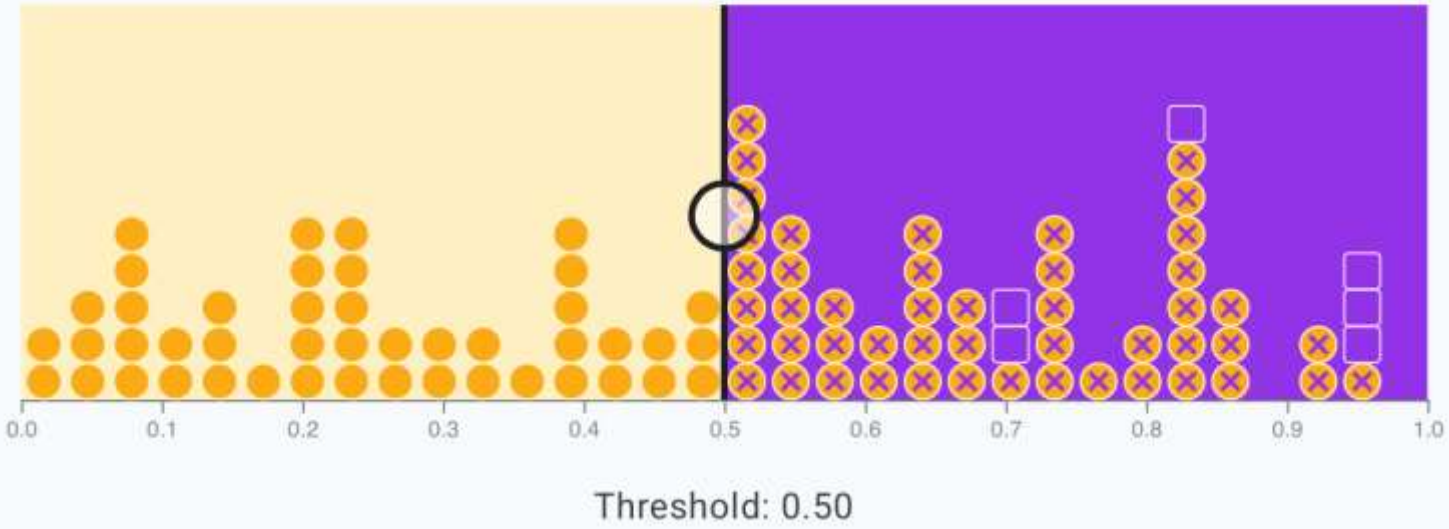
https://developers.google.com/machine-learning/crash-course/classification/thresholding

Confusion matrix

Dataset

Imbalanced

Classification threshold



Imbalanced dataset contains only a few examples of the positive class.

Confusion matrix

	Actually positive	Actually negative
Predicted positive	TP=6	FP=47
Predicted negative	FN=0	TN=46

https://developers.google.com/machine-learning/crash-course/classification/thresholding

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Evaluation metrics: Accuracy

Accuracy is the proportion of all classifications that were correct, whether positive or negative. It is mathematically defined as:

$$\text{Accuracy} = \text{correct classifications} / \text{total classifications} = TP + TN / TP + TN + FP + FN$$

In the spam classification example, accuracy measures the fraction of all emails correctly classified.

A perfect model would have zero false positives and zero false negatives and therefore an accuracy of 1.0, or 100%.

Accuracy incorporates all four outcomes from the confusion matrix (TP, FP, TN, FN), given a balanced dataset, with similar numbers of example in both classes, accuracy can serve as a coarse-grained measure of model quality.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Evaluation metrics: Accuracy

When the dataset is imbalanced, or where one kind of mistake (FN or FP) is more costly than the other, especially in most real-world applications, it's better to optimize for one of the other metrics instead.

For heavily imbalanced datasets, where one class appears very rarely, say 1% of the time, a model that predicts negative 100% of the time would score 99% on accuracy, despite being useless.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Evaluation metrics: Recall or true positive rate

The true positive rate (TPR), or proportion of all actual positives that were classified correctly as positives, is also known as recall.

Recall is mathematically defined as:

Recall (or TPR) = correctly classified actual positives / all actual positives = TP / TP + FN

False negatives are actual positives that were misclassified as negatives. That is why they appear in the denominator.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Evaluation metrics: Recall or true positive rate

A hypothetical perfect model would have zero false negatives and therefore a recall (TPR) of 1.0, which is to say, a 100% detection rate.

In an imbalanced dataset where the number of actual positives is very low, recall is a more meaningful metric than accuracy because it measures the ability of the model to correctly identify all positive instances.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Evaluation metrics: False positive rate

The false positive rate (FPR) is the proportion of all actual negatives that were classified incorrectly as positives, also known as the probability of false alarm. It is mathematically defined as:

FPR = incorrectly classified actual negatives / all actual negatives = FP / FP + TN

False positives are actual negatives that were misclassified, which is why they appear in the denominator.

A perfect model would have zero false positives and therefore a FPR of 0.0, which is to say, a 0% false alarm rate.

In a imbalanced dataset, where the number of actual negatives is very, very low, say 1-2 examples in total, FPR is less meaningful and less useful as a metric.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Evaluation metrics: Precision

Precision is the proportion of all the model's positive classifications that are actually positive. It is mathematically defined as:

Precision = correctly classified actual positives / everything classified as positive = TP / TP + FP

In the spam classification example, precision measures the fraction of emails classified as spam that were actually spam.

A hypothetical perfect model would have zero false positives and therefore a precision of 1.0.

In an imbalanced dataset where the number of actual positives is very, very low, say 1-2 examples in total, precision is less meaningful and less useful as a metric.

Precision improves as false positives decrease, while recall improves when false negatives decrease. As a result, precision and recall often show an inverse relationship, where improving one of them worsens the other.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Evaluation metrics: F1 score

The F1 score is the harmonic mean (a kind of average) of precision and recall.

Mathematically, it is given by:

F1 = 2 * (precision * recall / precision + recall) = (2TP / 2TP + FP + FN)

This metric balances the importance of precision and recall, and is preferable to accuracy for class-imbalanced datasets.

When precision and recall both have perfect scores of 1.0, F1 will also have a perfect score of 1.0.

More broadly, when precision and recall are close in values, F1 will be close to their value.

When precision and recall are far apart, F1 will be similar to whichever metric is worse.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Choice of metric and tradeoffs

The metric (s) you choose to prioritize when evaluating the model and choosing a threshold depend on the costs, benefits, and risks of the specific model.

Metric	Guidance
Accuracy	Use as a rough indicator of model training progress/convergence for balanced datasets. For model performance, use only in combination with other metrics. Avoid for imbalanced datasets. Consider using another metric.
Recall (True positive rate)	Use when false negatives are more expensive than false positives.
False positive rate	Use when false positives are more expensive than false negatives.
Precision	Use when it's very important for positive predictions to be accurate.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Topic 6: Model Evaluation

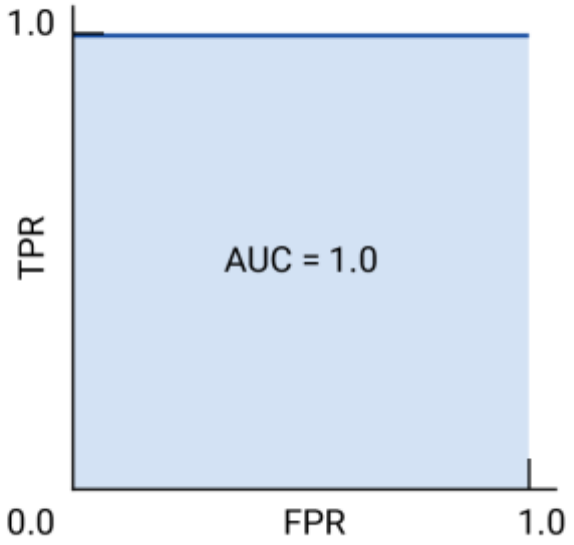
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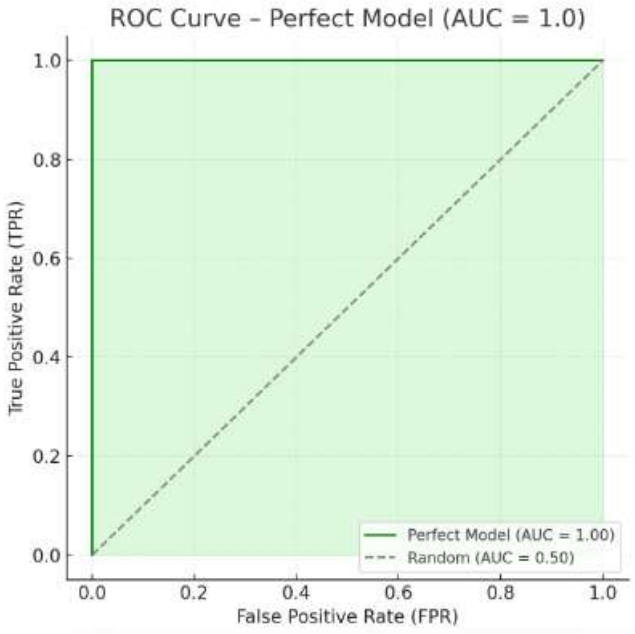
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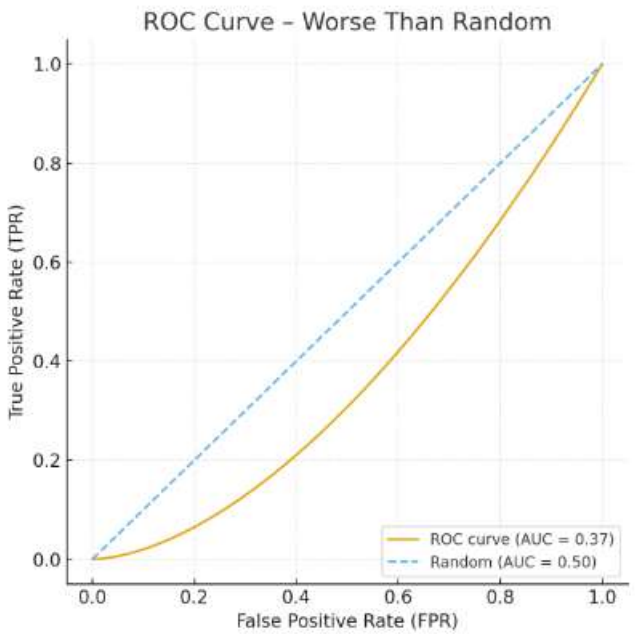
Receiver-operating characteristics curve (ROC)

The ROC curve is a visual representation of model performance across all thresholds. The long version of the name, receiver operating characteristic, is a holdover from WWII radar detection.

The ROC curve is drawn by calculating the true positive rate (TPR) and false positive rate (FPR) at every possible threshold, then graphing TPR over FPR. A perfect model, which at some threshold has a TPR of 1.0 and a FPR of 0.0, can be represented by either a point at (0, 1) if all other thresholds are ignored.





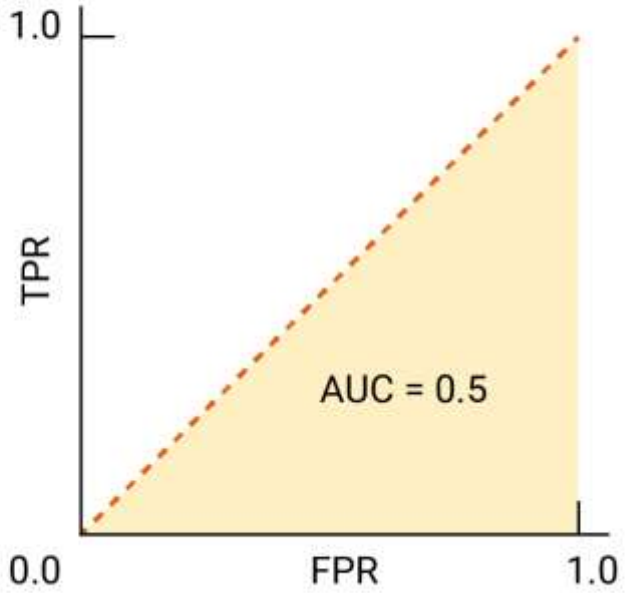


https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc

Area under the curve (AUC)

The AUC represents the probability that the model, of given a randomly chosen positive and negative example, will rank the positive higher than the negative.

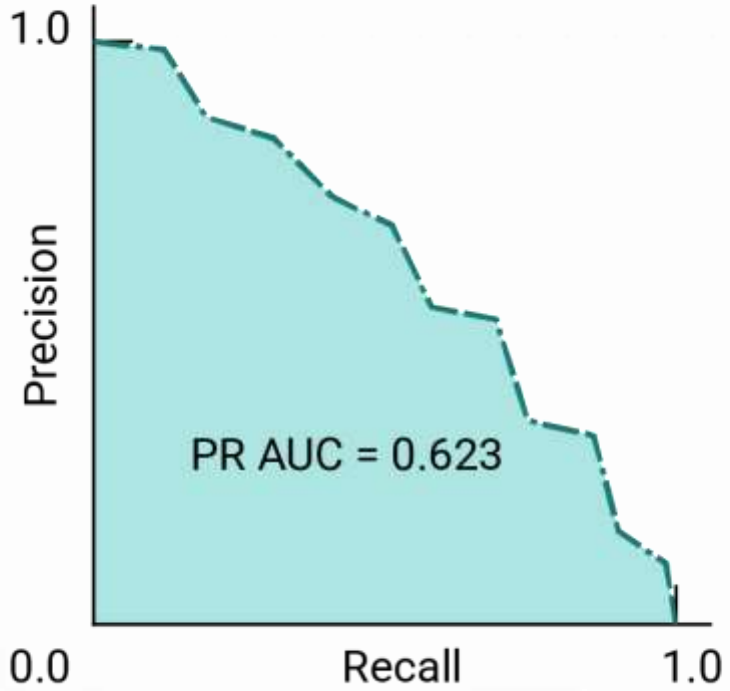
In the spam classifier example, a spam classifier with AUC of 0.5 assigns a random spam email a higher probability of being spam than a random legitimate email only half the time.



https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc

Precision-Recall curve (optional)

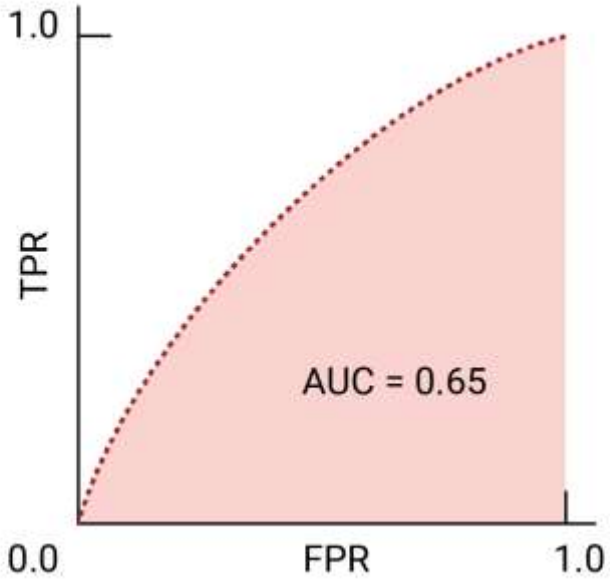
AUC and ROC work well for comparing models when the dataset is roughly balanced between classes. When the dataset is imbalanced, precision-recall curves (PRCs) and the area under those curves may offer a better comparative visualization of model performance. Precision-recall curves are created by plotting precision on the y-axis and recall on the x-axis across all thresholds.

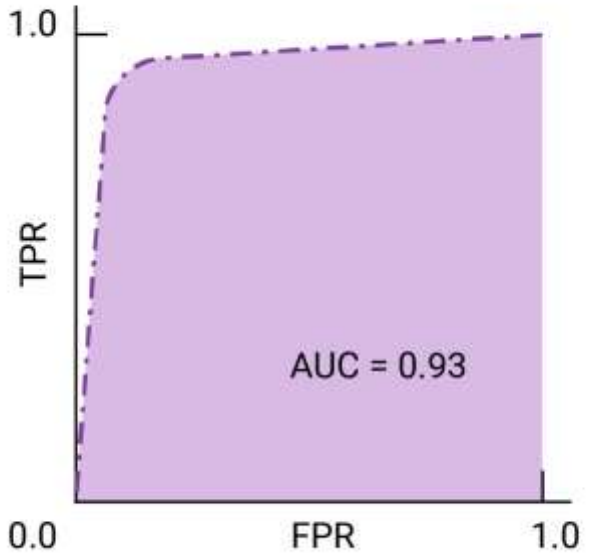


https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc

AUC and ROC for choosing model and threshold

AUC is a useful measure for comparing the performance of two different models, as long as the dataset is roughly balanced. The model with greater area under the curve is generally the better one.



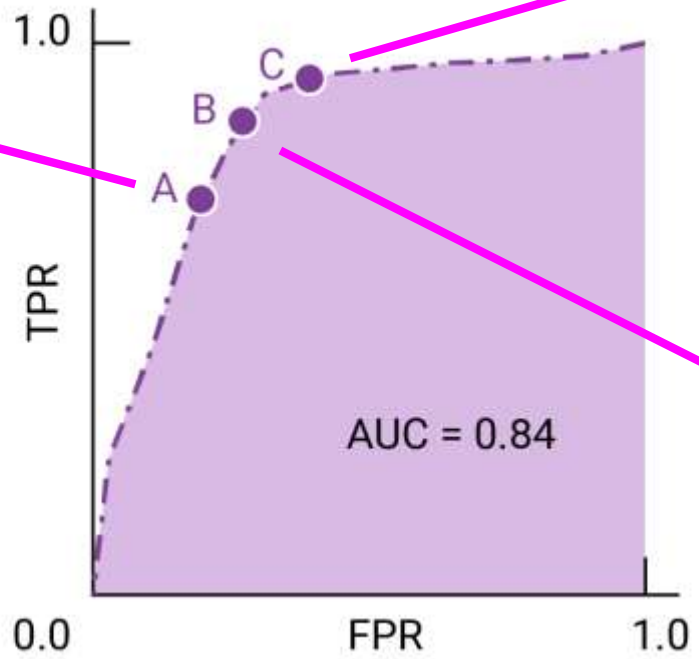


https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc

AUC and ROC for choosing model and threshold

The points on a ROC curve closest to (0,1) represent a range of the best-performing thresholds for the given model.

If false positives (false alarms) are highly costly, it may make sense to choose a threshold that gives a lower FPR, like the one at point A, even if TPR is reduced.



Conversely, if false positives are cheap and false negatives highly costly, the threshold for point C, which maximizes TPR, may be preferable.

If the costs are roughly equivalent, point B may offer the best balance between TPR and FPR.

https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc

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Prediction bias

Prediction bias is the difference between the mean of a model's predictions and the mean of ground-truth labels in the data.

Prediction bias can be caused by:

- Biases or noise in the data, including biased sampling for the training set.
- Too-strong regularization, meaning that the model was oversimplified and lost some necessary complexity.
- Bugs in the model training pipeline.
- The set of features provided to the model being insufficient for the task.

https://developers.google.com/machine-learning/crash-course/classification/prediction-bias

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Cross validation

Cross validation is a model assessment technique used to evaluate a machine learning algorithm's performance in making predictions on new datasets that it has not been trained on. This is done by partitioning the known dataset, using a subset to train the algorithm and the remaining data for testing.

Each round of cross-validation involves randomly partitioning the original dataset into a training set and a testing set. The training set is then used to train a supervised learning algorithm and the testing set is used to evaluate its performance. This process is repeated several times and the average cross-validation error is used as a performance indicator.

https://www.mathworks.com/discovery/cross-validation.html

Cross validation

When training a model, it is important not to overfit or underfit it with algorithms that are too complex or too simple. The training set and test set are critical in reducing this risk. However, dividing the dataset to maximize both learning and validity of test results is difficult.

Cross-validation also helps with choosing the best performing model by calculating the error using the testing dataset, which has not been used to train. The testing dataset helps to calculate the accuracy of the model and how it will generalize with future data.

https://www.mathworks.com/discovery/cross-validation.html

Cross validation

Many techniques are available for cross-validation. Among the most common are:

- **k-fold:** Partitions data into k randomly chosen subsets (or folds) of roughly equal size. One subset is used to validate the model trained using the remaining subsets. This process is repeated k times such that each subset is used exactly once for validation. The average error across all k partitions is reported as ε. This is one of the most popular techniques for cross-validation but can take a long time to execute because the model needs to be trained repeatedly.



https://www.mathworks.com/discovery/cross-validation.html

Cross validation

- **Holdout:** This method randomly splits the dataset into two parts, following a specific ratio, one for training and one for validation. It only runs training and testing once, which saves time on large datasets, but caution is advised when interpreting error metrics on smaller datasets.

- **Leaveout** involves dividing the dataset using a k-fold strategy where k equals the number of observations, ensuring each data point is used as the test set exactly once. This method is also referred to as leave-one-out cross-validation (LOOCV).

- **Stratify** divides the dataset ensuring that the training and test sets maintain approximately the same class distribution for the target variable.

https://www.mathworks.com/discovery/cross-validation.html

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Training, validation and test sets

A model should be tested against different set of examples than those used to train the model. Testing on different examples is stronger proof of your model's fitness than testing on the same set of examples. Where do you get those different examples? Traditionally in machine learning, you get those different examples by splitting the original dataset. You should slit the original dataset into two subsets:

- A training set that the model trains on.
- A test set for evaluation of the trained model.



Source: https://developers.google.com/machine-learning/crash-course/overfitting/dividing-datasets

Training, validation and test sets

Dividing the dataset into two sets is a decent idea, but a better approach is to divide the dataset into three subsets. In addition to the training set and the test set, the third subset is:

- A validation set performs the initial testing on the model as it is being trained.

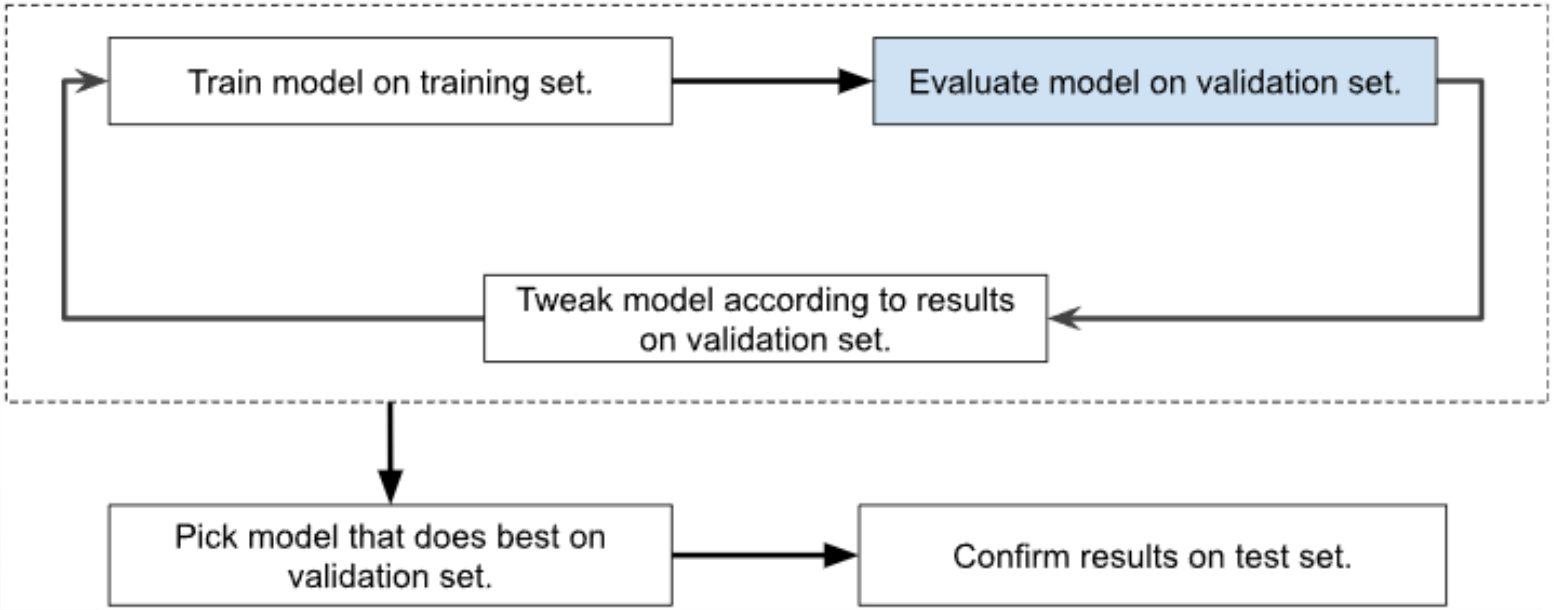


The validation set is used to evaluate results from the training set. After repeated use of the validation set suggests that your model is making good predictions, use the test set to double-check your model.

Source: https://developers.google.com/machine-learning/crash-course/overfitting/dividing-datasets

Training, validation and test sets

In the figure, “Tweak model” means adjusting anything about the model, from changing the learning rate, to adding or removing features, to designing a completely new model from scratch.



The workflow shown in Figure 10 is optimal, but even with that workflow, test tests and validation sets still “wear out” with repeated case. The more you use the same data to make decisions about hyperparameter settings or other model improvements, the less confidence that the model will make good predictions on new data. For this reason, it is good to collect more data to “refresh” the test set and validation set.

Source: https://developers.google.com/machine-learning/crash-course/overfitting/dividing-datasets

THANK YOU

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